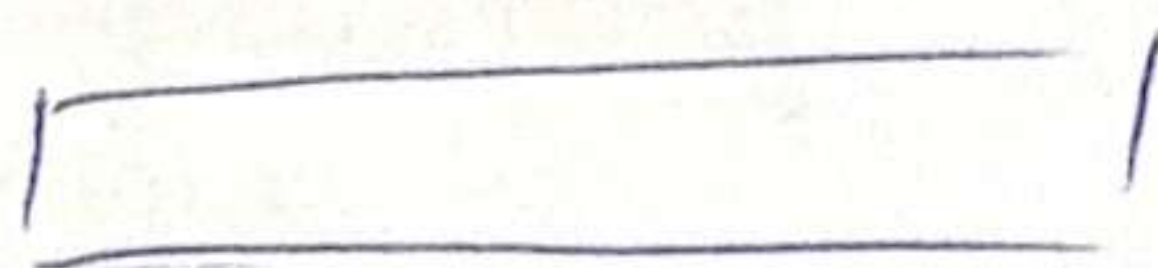


A.

DSA row 9

1. same as row 8/3

2. Stack on array
top = pos 1.



Rule: Array-based stack $\Rightarrow$ top at the end.

SLI-based: TOP at the head

DLI-based: TOP at either head / ^{tail} ~~thead~~

Problem: pushing an elem. needs linear time $\Rightarrow$ [b)

push() $\Rightarrow$ ^{new elem} must become new top $\Rightarrow$ insert at pos 1.
 $\Rightarrow$ all elems must shift one pos. to the right $\Rightarrow \Theta(n)$

pop(), top() $\Rightarrow \Theta(1)$ (they just remove pos 1)

3. 50 pos, collision res HT, sep. chaining

max. nr. of elems inserted $\Rightarrow$ [e) any nr

4. same as row 6

B. 2.1) same as row 6

2.2) same as row 6

2.3) a) $u-1$ — row 6

b) $1-1$ — row 6

c) min. elem. in max heap of N elems.

$\downarrow$
leaves

$\downarrow$
root

WC: $\Theta(N)$ we need to scan all leaves (not $\Theta(\text{height})$)

because one leaf is not enough to guarantee min

c. 3) $u-1$ — row 6

4) $u-u$ — row 6